



Module 05

MODEL SELECTION AND EVALUATION

Mohd Azraai Mohd Razman

mohdazraai@ump.edu.my

Anwar P.P. Abdul Majeed

amajeed@ump.edu.my

Lab Instructors

Muhammad Nur Aiman Shapiee

Lab Objectives

By the end of his lab, students should be able to understand the basic concept of Machine Learning, data analytic and the basic programming of machine learning in Orange Data Mining.

In this lab, you are required to have ample understanding on Machine Learning. The objectives of this lab exercises is to ensure students familiarise with Machine Learning algorithms and Orange Data Mining software. This lab sheet is structured as follows:

- i. PART A: Building Machine Learning Models
- ii. PART B: Improving the trained model.
- iii. PART C: Working on Regression Analysis
- iv. References

PART A: Building Machine Learning Models

Beginning with the basics and well-known model, we will first train a *logistic regression* model encompassing all the features just to understand how to select and build models.

Step 1: First, add a **Data Sampler** widget and connect it to **Data Table** widget.

Step 2: Add a **Logistic Regression** widget and **Test and Score** widget. Then, both widgets are connected to the **Data Sampler** widget. Details of widgets connection is demonstrated below:

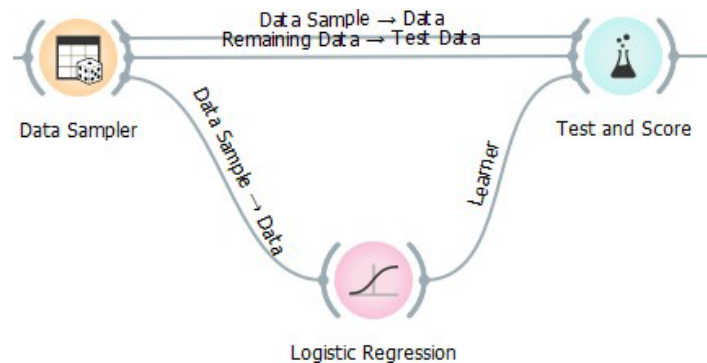


Figure 1 Workflow of building a model.

Step 3: Go to **Data Sampler** widget and double click on it. This widget is used to split the dataset into training set and test set.

Step 4: Choose the sampling type that we want. In this context, we set the sampling type to 'Fixed proportion of data' with 60%. The percentage indicates that 60% of data are used to train the model and the remaining data (40%) are used to test the model.

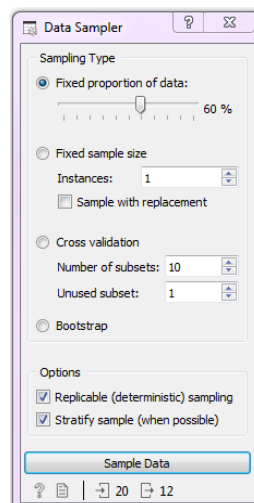


Figure 2 Data Sampler

Step 5: Double click on **Logistic Regression** widget. Few parameters are shown, and these parameters can be manipulated in order to obtain good results. But for basic and easy understanding, we encouraged that do not change any of the settings. Let the model be in default setting.

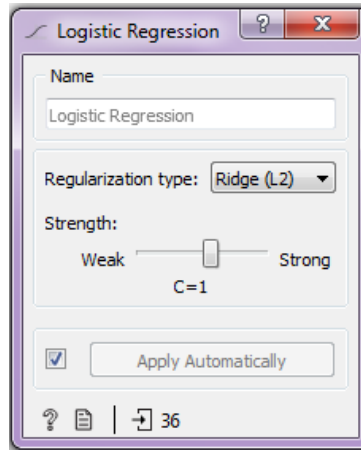


Figure 3 Logistic Regression

Step 6: Double click on **Test and Score** widget. For training the model, first we select the 'Leave one out' option (red box). The selection of this method is because of the dataset is small. The score of the trained model was illustrated (orange box).

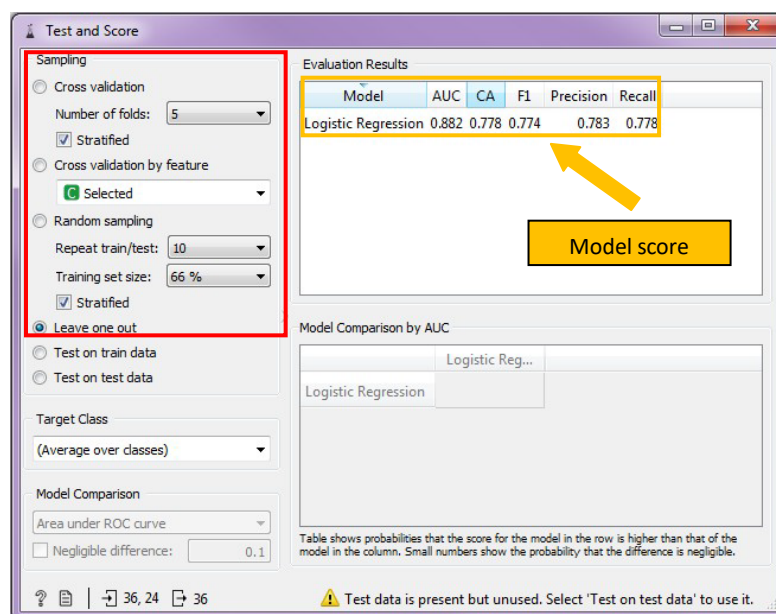


Figure 4 Score of trained model

To validate the trained model, click on 'Test on test data' option (red box). The final results of the model are shown (orange box).

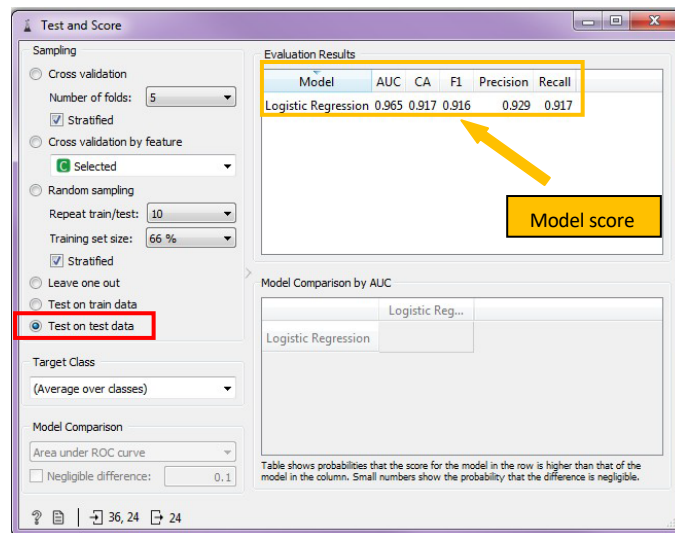


Figure 5 Test score on trained model

Step 7: Add a **Confusion Matrix** widget and connect it to **Test and Score** widget. The use of **Confusion Matrix** widget is to visualize the results better.

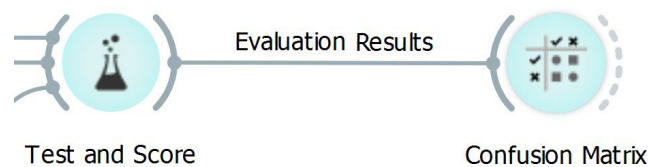


Figure 6 Test and Score linked with Confusion Matrix

Step 8: Double click on **Confusion Matrix** widget. As we can see, in training and testing the model, there are few classes that misclassified (highlighted in red) and data that correctly classified highlighted in blue.

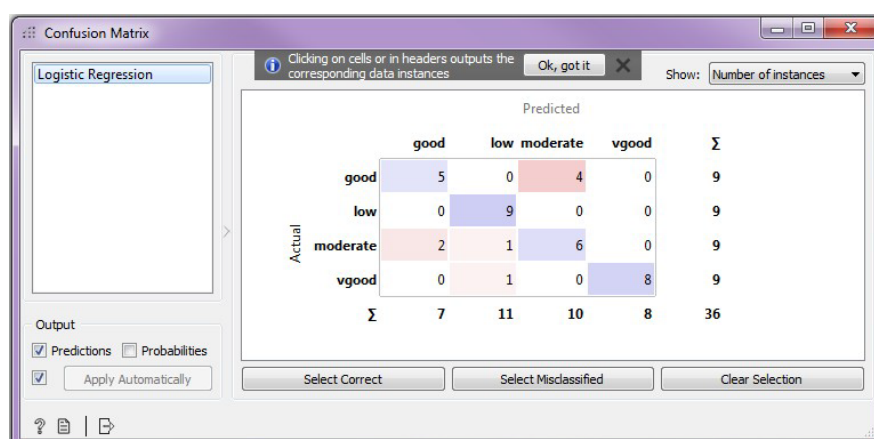


Figure 7 Confusion Matrix of trained Logistic Regression

For testing the model, only two data were misclassified.

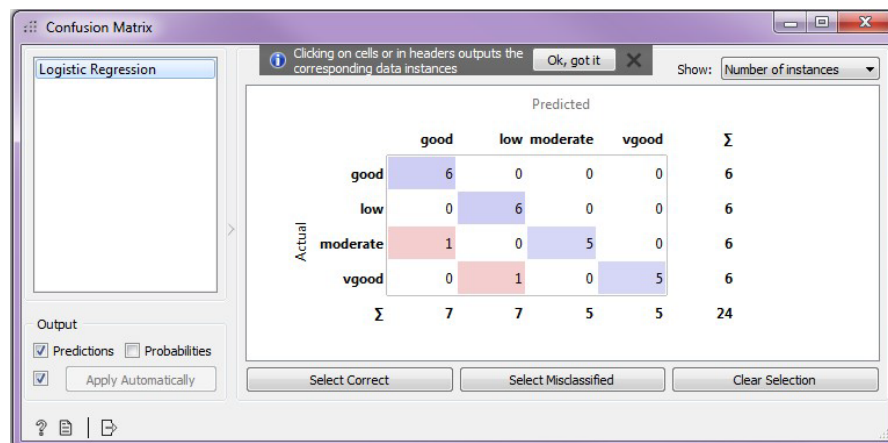


Figure 8 Confusion Matrix of testing Logistic Regression

This way, you can test out different models and see how accurately they perform.

Let's try to evaluate how a Support Vector Machine, Neural Network, and k-Nearest Neighbors would do? Change the modeling method to Support Vector Machine, Neural Network, and k-Nearest Neighbors and look at the confusion matrix. Figure 9 and 10 illustrated the train and test results for multiple models.

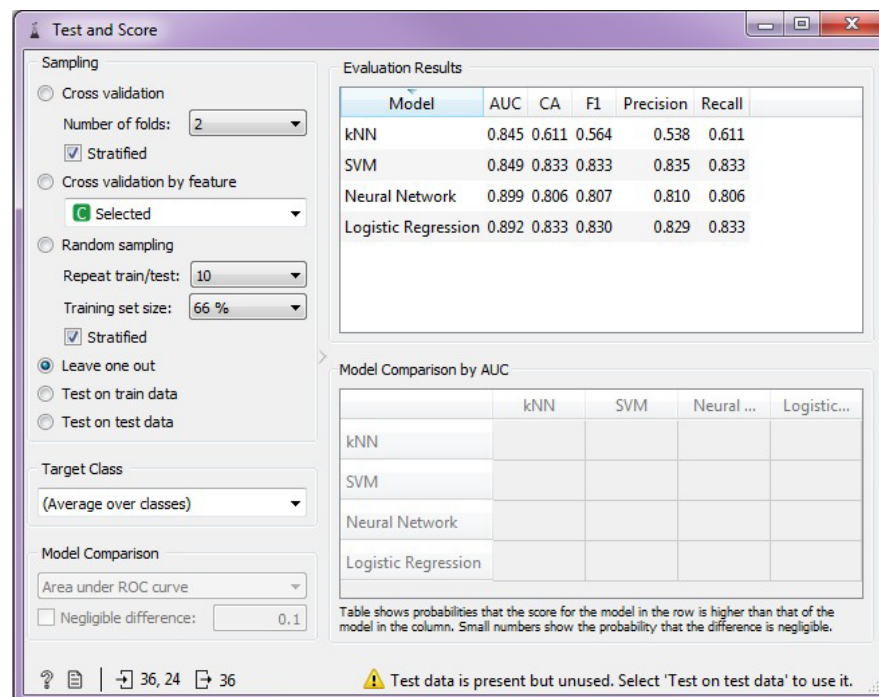


Figure 9 Training results for four classifiers

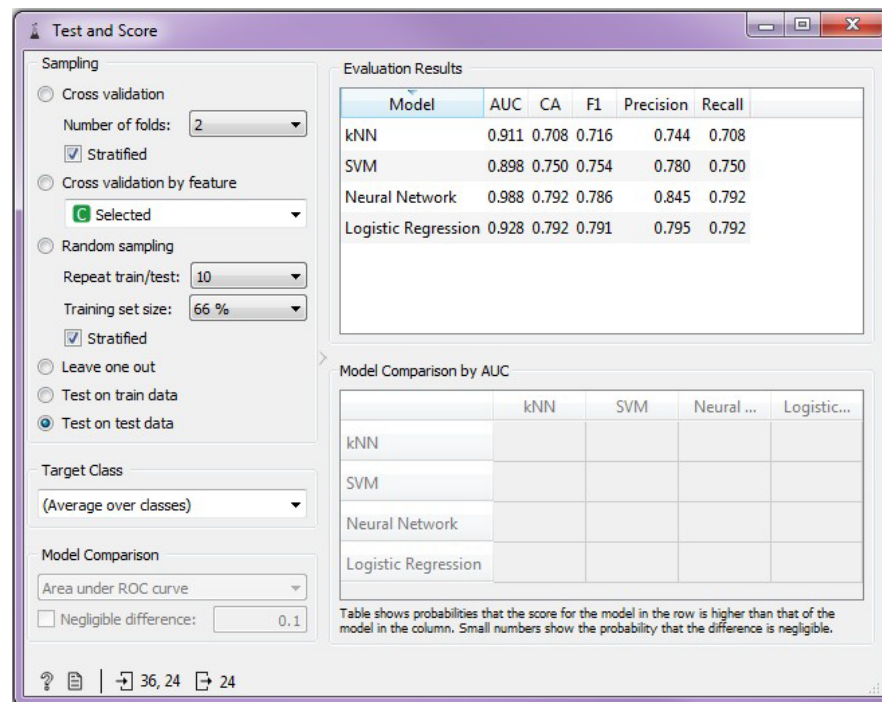


Figure 10 Test results for four classifiers

Step 9: We can save the developed model by adding **Save Model** widget and connect to our trained model. Then, double click the **Save Model** and click “Save as...” to save the model. For example, see below.

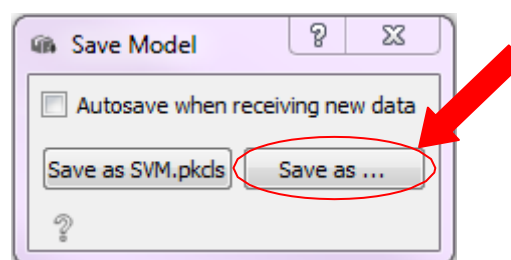


Figure 11 Save model workflow

Step 10: The saved model can be load for further analysis by adding **Load Model** widget and connect to **Data Sampler** and **Predictions** widgets. Then, double click the **Load Model** and click “file icon” to browse and load the model. For example, see below.

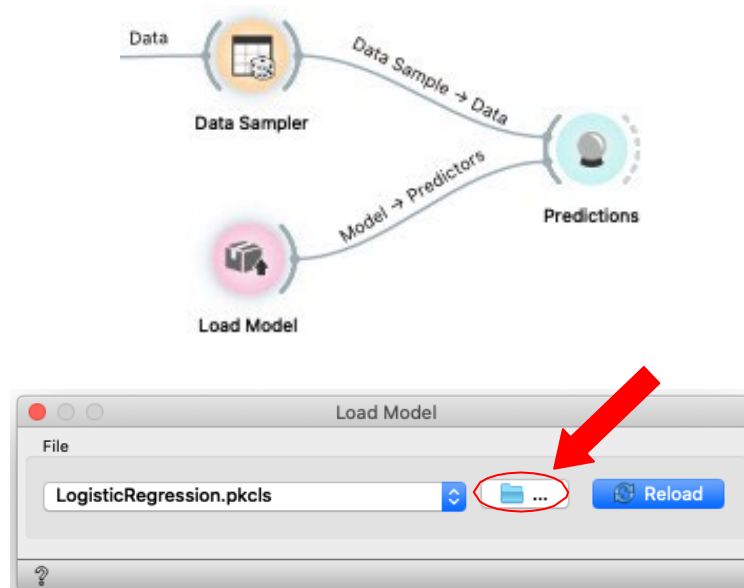


Figure 12 Load model workflow

Step 11: Double click on **Predictions** widget to see the performance of the test set.

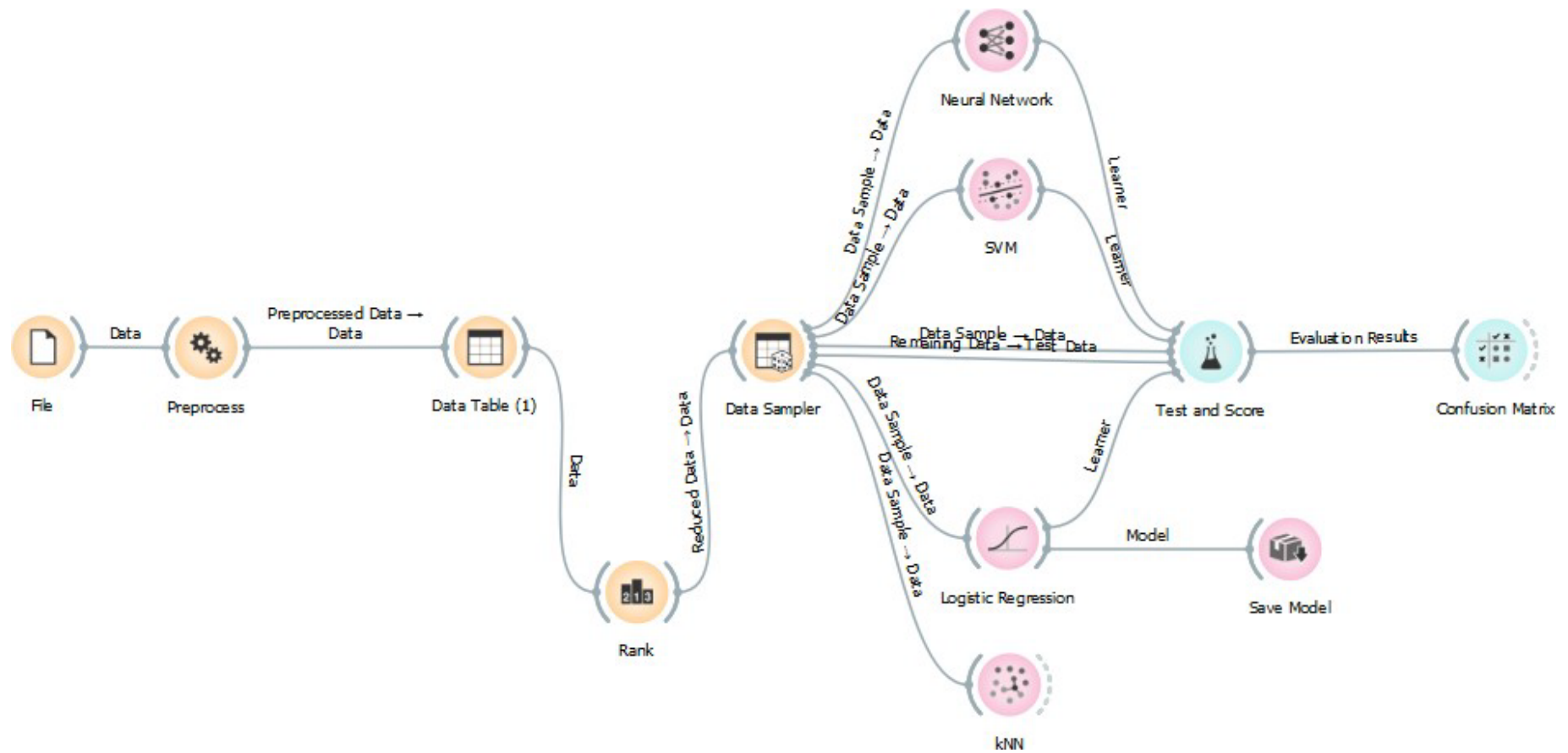
The screenshot shows the 'Predictions' widget interface. On the left, there's a 'Show probabilities for' section with a list: 'good', 'low', 'moderate', 'vgood'. The main area displays a table of results for 25 samples. The table has columns for 'Quality', 'Doors', 'Selected', 'Persons', and 'Maintenance'. Below the table is a summary table with columns: 'Model', 'AUC', 'CA', 'F1', 'Precision', and 'Recall'.

Quality	Doors	Selected	Persons	Maintenance
moderate	3.0	No	more	vhhigh
vgood	5more	No	more	high
vgood	2.0	No	more	high
vgood	4.0	No	?	high
low	3.0	No	4.0	med
moderate	2.0	No	more	vhhigh
vgood	4.0	No	more	high
good	2.0	No	4.0	vhhigh
moderate	3.0	No	4.0	vhhigh
moderate	3.0	No	4.0	vhhigh
low	4.0	No	2.0	?
moderate	2.0	No	more	vhhigh
low	4.0	No	2.0	med
good	3.0	No	2.0	vhhigh
low	3.0	No	more	med
vgood	4.0	No	4.0	med
good	2.0	No	4.0	med

Model	AUC	CA	F1	Precision	Recall
Logistic Regression	0.972	0.905	0.901	0.904	0.905

Figure 13 Predictions widget

System workflow overview



Quick Exercise 1 (5 marks)

- 1. In which of the following scenarios would a high accuracy value suggest that the ML model is doing a good job?**
 - a) A deadly, but curable, medical condition afflicts .01% of the population. An ML model uses symptoms as features and predicts this affliction with an accuracy of 99.99%.
 - b) An expensive robotic chicken crosses a very busy road a thousand times per day. An ML model evaluates traffic patterns and predicts when this chicken can safely cross the street with an accuracy of 99.99%.
 - c) In the game of roulette, a ball is dropped on a spinning wheel and eventually lands in one of 38 slots. Using visual features (the spin of the ball, the position of the wheel when the ball was dropped, the height of the ball over the wheel), an ML model can predict the slot that the ball will land in with an accuracy of 4%.
- 2. Consider a classification model that separates email into two categories: "spam" or "not spam." If you raise the classification threshold, what will happen to precision?**
 - a) Probably increase.
 - b) Definitely decrease.
 - c) Definitely increase.
 - d) Probably decrease.

Question 2 (15 marks)

The previous tutorial evaluated the trained model against the training set, which does not provide a strong signal about the quality of your model. In this problem, you'll experiment with train sets, validation sets and test sets. Given dataset on Vehicle Silhouette to predict vehicle type based on silhouette measurements.

Learning objectives

- Split the dataset into 70% for train sets and 30% for test sets.
- Use multiple machine learning models to evaluate the datasets.
- Test the trained model with a test set to determine whether your trained model is overfitting.
- Try to reduce the features to obtain a better result and choose the best machine learning model.

Case Study (30 marks)

A Chinese automobile company Geely Auto aspires to enter the US market by setting up their manufacturing unit there and producing cars locally to give competition to their US and European counterparts. They have contracted an automobile consulting company to understand the factors on which the pricing of cars depends. Specifically, they want to understand the factors affecting the pricing of cars in the American market, since those may be very different from the Chinese market.

Using the given dataset on Car Pricing in American Market, you are required to model the price of cars with the available independent variables. It will be used by the management to understand how exactly the prices vary with the independent variables. They can accordingly manipulate the design of the cars, the business strategy etc. to meet certain price levels. Further, the model will be a good way for management to understand the pricing dynamics of a new market.

Learning objectives

- Split the dataset into 70% for train sets and 30% for test sets.
- Find the significant variables in predicting the price of a car
- Use multiple regression models to evaluate the datasets.
- Test the trained model with a test set to determine how well those variables describe the price of a car.

REFERENCES

Medium, Introduction to machine learning
Analytics Vidhya, Building Machine Learning